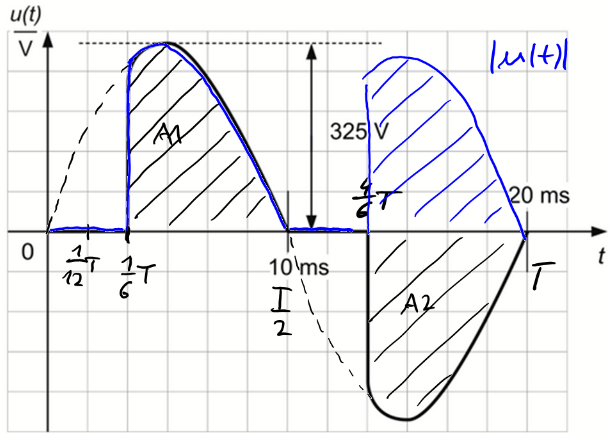
PER 4 (Klausur SS 1996)

Gegeben ist folgender periodischer Spannungsverlauf:



$$u(t) = \hat{u} \sin(\omega t) \qquad \hat{u} = 325V$$

a) Mittelwert

$$\bar{u} = \frac{1}{T}\int_0^T u(t)\,dt = 0$$ aus Symmetriegründen (A1 = A2)

b) Gleichrichtwert

$$\overline{|u|} = \frac{1}{T}\int_0^T |u(t)|\,dt$$

$$= \frac{1}{\frac{T}{2}} \int_{\frac{1}{6}T}^{\frac{T}{2}} \hat{u} \sin(\omega t)\,dt$$

$$= \frac{2}{T}\,\hat{u}\,\left(\frac{-\cos(\omega t)}{\omega}\right)\Bigg|_{\frac{1}{6}T}^{\frac{T}{2}}$$

$$= \frac{2}{\omega T}\,\hat{u}\left[-\cos\left(\omega\cdot\frac{T}{2}\right) + \cos\left(\omega\cdot\frac{T}{6}\right)\right]$$

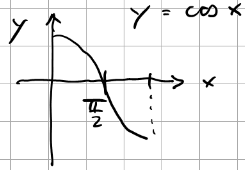
Nebenrechnung:

$$\omega T = 2\pi f \cdot T \qquad f = \frac{1}{T}$$

$$= 2\pi \frac{1}{\cancel{T}} \cdot \cancel{T}$$

$$= 2\pi$$

$$= \frac{2}{2\pi} \hat{u} \left[-\cos\left(\frac{2\pi}{2}\right) + \cos\left(\frac{2\pi}{6}\right) \right]$$



$$= \frac{\hat{u}}{\pi} \left[-(-1) + \frac{1}{2} \right]$$

$$= \frac{3}{2} \frac{\hat{u}}{\pi} = \frac{3}{2} \cdot \frac{325V}{\pi}$$

$$= 155{,}2\,V$$

c)

$$U_{RMS} = \sqrt{\frac{1}{T} \int_0^T u^2(t)\,dt}$$

$$= \sqrt{\frac{1}{\frac{T}{2}} \int_{\frac{1}{6}T}^{\frac{T}{2}} (\hat{u} \sin(\omega t))^2\,dt}$$

$$= \sqrt{\frac{2}{T} \hat{u}^2 \int_{\frac{1}{6}T}^{\frac{T}{2}} \sin^2(\omega t)\,dt}$$

Mathe: $\sin^2 x = \frac{1}{2} [1 - \cos 2x]$

$$U_{RMS} = \sqrt{\frac{2}{T} \hat{u}^2 \int_{\frac{1}{6}T}^{\frac{T}{2}} \frac{1}{2} [1 - \cos(2\omega t)]\,dt}$$

$$= \sqrt{\frac{2}{T} \hat{u}^2 \frac{1}{2} \left[t - \frac{\sin(2\omega t)}{2\omega} \right]_{\frac{T}{6}}^{\frac{T}{2}}}$$

$$= \sqrt{\frac{\hat{u}^2}{T} \left[\frac{T}{2} - \frac{\sin(2\omega \frac{T}{2})}{2\omega} - \frac{T}{6} + \frac{\sin(2\omega \cdot \frac{T}{6})}{2\omega} \right]}$$

$$= \sqrt{\frac{\hat{u}^2}{T} \left[\frac{T}{3} - \frac{\sin(\omega T)}{2\omega} + \frac{\sin(\frac{1}{3}\omega T)}{2\omega} \right]}$$

$$= \sqrt{\frac{\hat{u}^2}{T}\left[\frac{T}{3} - \frac{\cancel{\sin(2\pi)}^{\,0}}{2\omega} + \frac{\cancel{\sin\left(\frac{2}{3}\pi\right)}^{\,\frac{1}{2}\sqrt{3}}}{2\omega}\right]}$$

mit $\omega T = 2\pi$

$$= \sqrt{\frac{\hat{u}^2}{T}\left[\frac{T}{3} + \frac{1}{2}\frac{\sqrt{3}}{2\omega}\right]}$$

mit $\omega = 2\pi f = 2\pi \cdot \frac{1}{T}$

$$= \sqrt{\frac{\hat{u}^2}{\cancel{T}}\left[\frac{\cancel{T}}{3} + \frac{1}{2}\frac{\sqrt{3}\,\cancel{T}}{2\cdot 2\pi}\right]}$$

$$= \hat{u}\sqrt{\frac{1}{3} + \frac{\sqrt{3}}{8\pi}}$$

$$= 325\,V\sqrt{\frac{1}{3} + \frac{\sqrt{3}}{8\pi}}$$

$$= \underline{\underline{206{,}1\,V}}$$